Control of coupled penduli

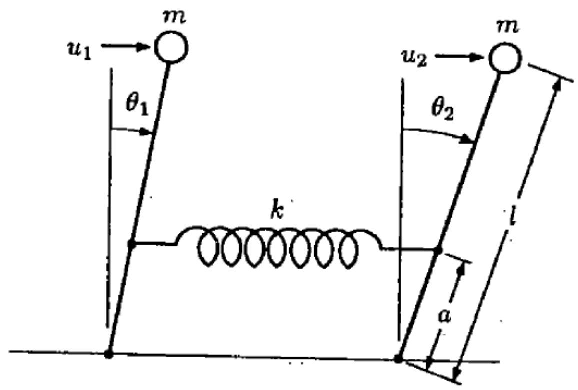

Fig. 1.16. Inverted penduli.

1.36. EXAMPLE. To illustrate a *physical* decomposition of a dynamic system, let us consider two identical penduli that are coupled by a spring and subject to two distinct inputs as shown in Figure 1.16. The equations of motion are

$$\begin{aligned} m\ell^2\ddot{\theta}_1 &= mg\ell\theta_1 - ka^2(\theta_1 - \theta_2) + u_1, \qquad & y_1 &= \theta_1, \\ m\ell^2\ddot{\theta}_2 &= mg\ell\theta_2 - ka^2(\theta_2 - \theta_1) + u_2, \qquad & y_2 &= \theta_2. \end{aligned} \tag{1.61}$$

Choosing the state vector as $x = (\theta_1,\, \dot{\theta}_1,\, \theta_2,\, \dot{\theta}_2)$ and denoting $u = (u_1,\, u_2)^T$, (1.61) can be rewritten as

$$\mathbf{S}\colon\; \dot{x} = \left[\begin{array}{cc:cc} 0 & 1 & 0 & 0 \\ \frac{g}{\ell} - \frac{ka^2}{m\ell^2} & 0 & \frac{ka^2}{m\ell^2} & 0 \\ \hdashline 0 & 0 & 0 & 1 \\ \frac{ka^2}{m\ell^2} & 0 & \frac{g}{\ell} - \frac{ka^2}{m\ell^2} & 0 \end{array}\right] x + \left[\begin{array}{c:c} 0 & 0 \\ \frac{1}{m\ell^2} & 0 \\ \hdashline 0 & 0 \\ 0 & \frac{1}{m\ell^2} \end{array}\right] u,$$

resulting in the system "centralized" model having the form

$$\dot{x} = Ax + Bu$$

where the matrices A and B given in the corresponding MATLAB file. We assume that all the state variables are measurable.

Problem:

Perform the following tasks considering the following values of parameter k:

a) $k = 0.02$ N/m;
b) $k = 2$ N/m;
c) $k = 200$ N/m.

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time $h = 0.5$ s). Perform the following analysis:
    a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
    b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions:
    a. Compute the continuous-time fixed modes
    b. Compute the discrete-time fixed modes
    c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
    d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
    e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the angles θ_1 and θ_2 starting from a common random initial condition.